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PubMed Digest - 2026-10-28

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PubMed Digest - airborne transmission of respiratory viruses (papers added 2026-10-22 to 2026-10-28)

Note: no earlier digests were found in sent mail, so no papers were deduplicated. Summaries are based on the abstract text in the feed, which is truncated for some items.

1. Predictive Performance of Inflammatory Markers for Severe Disease Progression in Measles: A Retrospective Cohort Analysis
Retrospective analysis of 509 hospitalized measles patients in a Romanian outbreak, testing how well routine admission inflammatory biomarkers predict severe disease and prolonged stay.
https://www.google.com/url?q=https://pubmed.ncbi.nlm.nih.gov/42943553/&source=gmail&ust=1793292030366000&sa=E

2. An Autonomous Pathogen Detection System for Assessing Temporal Associations with Respiratory Pathogen Activity
Describes an autonomous system combining real-time bioaerosol sensing, triggered aerosol collection, nucleic acid extraction, real-time PCR and automated reporting, tested in a 20 m3 chamber and over 52 weeks of continuous monitoring.
https://www.google.com/url?q=https://pubmed.ncbi.nlm.nih.gov/42943430/&source=gmail&ust=1793292030366000&sa=E

3. A Thermophilic Bacteriophage TP-84 Surrogate System for Validating Photocatalytic and Ultraviolet Inactivation of Airborne Viruses, Including SARS-CoV-2
Validates phage TP-84 as a BSL-1 surrogate challenge organism for testing TiO2 photocatalytic and UV air-disinfection controls against airborne viruses.
https://www.google.com/url?q=https://pubmed.ncbi.nlm.nih.gov/42942040/&source=gmail&ust=1793292030366000&sa=E

4. Photoactive Chiral Nanoneedles for Site-Selective Cleavage and Broad-Spectrum Inactivation of Viruses
Chiral ZnO nanoneedle coatings that capture and inactivate viruses by targeting the HR2 region of class I fusion proteins, aimed at making masks and filters inactivate rather than just trap pathogens.
https://www.google.com/url?q=https://pubmed.ncbi.nlm.nih.gov/42937551/&source=gmail&ust=1793292030366000&sa=E

5. Influenza A (H1N1) and SARS-CoV-2 Transmission in Airplanes: Quanta Profiles and Infection Control
Estimates population-level quanta generation rate profiles for H1N1pdm09 and ancestral SARS-CoV-2 to support airborne infection risk assessment in aircraft and other high-occupancy spaces, accounting for short-range transmission.
https://www.google.com/url?q=https://pubmed.ncbi.nlm.nih.gov/42937471/&source=gmail&ust=1793292030366000&sa=E

6. Germicidal ultraviolet to reduce transmission of respiratory infections indoors: a systematic review
Concludes that older clinical studies suggest germicidal UV reduces highly airborne-transmissible infections, but broader effectiveness is uncertain; inconclusive results may reflect methodological limits, and adequately powered RCTs are called for.
https://www.google.com/url?q=https://pubmed.ncbi.nlm.nih.gov/42929067/&source=gmail&ust=1793292030366000&sa=E

7. Combating Infectious Disease Challenges: Have We Gone Twenty Steps Forward or Backward?
Write-up of a 2025 webinar arguing for a public-health paradigm shift on airborne pathogens, noting continued reliance on surgical masks and on the assumption that aerosol-generating procedures are needed for spread.
https://www.google.com/url?q=https://pubmed.ncbi.nlm.nih.gov/42925011/&source=gmail&ust=1793292030366000&sa=E

8. A Superposition Model of Droplet Exposure to SARS-CoV-2
Combines data from five published studies to estimate relative virion exposure from short- and long-range airborne transmission at 1 m and beyond, using an 8 um droplet size for comparison.
https://www.google.com/url?q=https://pubmed.ncbi.nlm.nih.gov/42921593/&source=gmail&ust=1793292030366000&sa=E

Relevance to your background

- Sensing and measurement: The autonomous detection system (42943430) is a full sensing-to-PCR pipeline with a 52-week field deployment, a useful reference for triggering logic, sampling design and validating bioaerosol monitors against chamber tests.
- Ventilation, filtration and disinfection engineering: The GUV systematic review (42929067), the TP-84 surrogate paper (42942040) and the nanoneedle coating work (42937551) together cover the engineering-controls side. The review's point that inconclusive results may stem from methodology matters for designing evaluations, and a BSL-1 surrogate makes bench testing of UV/photocatalytic units more practical.
- Modelling and simulation: The aircraft quanta-profile paper (42937471) and the superposition exposure model (42921593) are both exposure-modelling work: Wells-Riley-style quanta estimation, and combining near-field and far-field contributions. Both bear on the assumptions you would carry into a risk or CFD model.
- Study design and data quality: The GUV review's call for adequately powered RCTs with robust endpoints and the superposition model's reliance on pooled data from five studies (with a single fixed droplet size) are worth reading critically for heterogeneity and confounding. The measles biomarker paper (42943553) is a clinical cohort study and only loosely related to the airborne topic.